Pro Football Sports Science Research Update

November 2025

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Adulteration of sports supplements with anabolic steroids - From innocent athlete to vicious cheater. Nutrients, 17(19): 3146, 2025.

"Can We Trust Them?"- An Expert Evaluation of Large Language Models to Provide Sleep and Jet Lag Recommendations for Athletes. Sports Medicine, Sports Medicine, Ahead of Print: 1-14, 2025.

Characterizing body postures by injury scenarios - A video review analysis of hamstring strain injuries in the National Football League. BMJ Open Sport & Exercise Medicine, 11(4): e002749, 2025.

Club-based vs hospital-guided rehabilitation after ACL reconstruction in football participants: A two-year prospective cohort study. Frontiers in Sports and Active Living, 7: 1641750, 2025.

Commotio cordis in sudden cardiac death in the young: A state-of-the-art-review. Reviews in Cardiovascular Medicine, 26(9): 43357, 2025.

Considering the effects of cannabinoids and exercise on the brain: A narrative review. Sports, 13: 320, 2025.

Does the warmup really reduce injury risk? Why we need new approaches to confidently answer this question. Journal of Science and Medicine in Sport, Ahead of Print: 1-3, 2025.

Dose-response relationship between evening exercise and sleep. Nature Communications, 16: 3297, 2025.

Effects of post-exercise heat exposure on acute recovery and training-induced performance adaptations: A systematic review. Sports Medicine – Open, 11: 106, 2025.

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Ergogenic and physiological effects of sports supplements: Implications for advertising and consumer information. Nutrients, 17: 2706, 2025.

Foot strike angle as key predictor of multi-plane knee joint moments in sidestep cutting: Implications for ACL injury prevention. Sports Biomechanics, Ahead of Print: 1-12, 2025.

Hamstring injury risk in male professional football (soccer): Do external training loads play a role? BMJ Open Sport & Exercise Medicine, 11: e002649, 2025.

Impact of ankle injury history and pre-National Football League Draft training on lower limb coordination during running: A pilot study. Cureus, 17(10): e93887, 2025.

Impact of evening blue light exposure timing on sleep, motor, and cognitive performance in young athletes with intermediate chronotype. Biology of Sport, 42(3): 61-68, 2025.

Implementing exposure limits for players in contact team sports: Review of principles and practices. Frontiers in Sports and Active Living, 7: 1593766, 2025.

Is it all about the stars? Spending on star players and success in the National Football League. Omega: The International Journal of Management Science, Ahead of Print: 1-31, 2025.

Kinetics and kinematics of initial sprint acceleration performance in Division 1 collegiate American football players. Journal of Sports Science, Ahead of Print: 1-15, 2025.

Mediterranean diet and athletic performance in elite and competitive athletes: A systematic review and meta-analysis. Nutrition, Metabolism and Cardiovascular Diseases, Ahead of Print: 1-15, 2025.

Mental fatigue negatively impacts cognitive and physical performance outcomes: A large-scale randomized crossover trial. Medicine and Science in Sports and Exercise, Ahead of Print: 1-4687, 2025.

Motion capture technologies for athletic performance enhancement and injury risk assessment: A review for multi-sport organizations. Sensors, 25: 4384, 2025.

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Pro Football Sports Science Research Update

Next-Generation Approaches in Sports Medicine: The Role of Genetics, Omics, and Digital Health in Optimizing Athlete Performance and Longevity—A Narrative Review. Life, 15: 1023, 2025.

Peri-operative protein or amino acid supplementation for total joint arthroplasty: A systematic review and meta-analysis. Journal of Orthopaedic Surgery and Research, 20: 439, 2025.

Player-Level Tackle Training Interventions in Tackle-Collision Sports: A Systematic Scoping Review. Sports Medicine – Open, 11: 103, 2025.

Practices and perceptions in hamstring training for injury prevention and enhancement of athletic performance: A qualitative analysis. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2025.

Proprioception and Central Nervous Excitability: Implications for Return-to-Play in Athletes with Ligament or Meniscal Injuries. International Journal of Sports Physical Therapy, 20(8): 1122-1130, 2025.

Redefining the draft pick valuation in the National Football League. Frontiers in Sports and Active Living, 7: 1628223, 2025.

Rethinking Acute Sports Injuries: Evidence for an Overuse Mechanism in Hamstring and ACL Injuries. Scandinavian Journal of Medicine & Science in Sports, 35: e70146, 2025.

The Dose–Response Relationship of Exercise-Based Injury Prevention Programs: Implications for Research and Practice. Sports Medicine, Ahead of Print: 1-10, 2025.

The Impact of Injury Severity on Sleep Quality in Football (Soccer) Players. European Journal of Sport Science, 4(3): 1-8, 2025.

The Role of Vitamin D in Postoperative Tendon Healing: A Scoping Review. The Orthopaedic Journal of Sports Medicine, 13(10): 1-9, 2025.

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Uphill Walking Unveils Hidden Instability and Biomechanical Adaptations in Anterior Cruciate Ligament-Deficient Copers. Gait & Posture, Ahead of Print: 1-26, 2025.

A deep learning algorithm for automatic 3D segmentation and quantification of hamstrings musculotendon injury from MRI. Scientific Reports, 15(1): 33497, 2025.

Abstract - In high-velocity sports, hamstring strain injuries are common causes of missed play and have high rates of reinjury. Evaluating the severity and location of a hamstring strain injury, currently graded by a clinician using a semiqualitative muscle injury classification score (e.g. as one method, British Athletics Muscle Injury Classification - BAMIC) to describe edema presence and location, aids in guiding athlete recovery. In this study, automated artificial intelligence (AI) models were developed and deployed to automatically segment edema, hamstring muscle and tendon structures using T2-weighted and T1-weighted magnetic resonance images (MRI), respectively. MR scans were collected from collegiate football athletes at time-of-hamstring injury and return to sport. Volume, length, and cross-sectional (CSA) measurements were performed on all structures and subregions (i.e. free tendon and aponeurosis). The edema and hamstring muscle/tendon AI models compared favorably with ground-truth segmentations. AI volumetric output correlated with ground truth for edema ($R = 0.97$), hamstring muscles ($R \geq 0.99$), and hamstring tendon ($R \geq 0.42$) structures. Edema volume and percentage of muscle impacted by edema significantly increased with clinical BAMIC grade ($p < 0.05$). Taken together, these results demonstrate a promising new approach for AI-based quantification of edema which reflects differing levels of injury severity and supports clinical validity.

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Adulteration of sports supplements with anabolic steroids - From innocent athlete to vicious cheater. Nutrients, 17(19): 3146, 2025.

Abstract - Some protein food supplements intended for athletes may be adulterated with pharmacologically active substances, including anabolic steroids and prohormones. The addition of these substances is aimed at enabling manufacturers to achieve rapid sales growth by promising quick increases in strength and muscle mass. However, the consumption of these products will lead to a positive result in a routine anti-doping test, along with all of the consequences that will directly affect an athlete's career and reputation. At the same time, the illicit use of anabolic steroids continues to evolve across numerous sport disciplines. Moreover, vicious cheaters try to cover up their illegal actions by using various pharmacological agents to mask detection in anti-doping tests. This narrative review focuses on two situations-the innocent athlete and the vicious cheater. The athlete involved in inadvertent doping will suffer the consequences of doping, making close collaboration with medical staff extremely important. The analytic strategies described here address anabolic steroid doping detection and cheating using masking agents. This approach, based on biochemical changes, examines how these substances interfere with the testosterone pathway, from synthesis to elimination. Using masking agents alters the steroid profile, and the modifications produced by each agent are the subject of a detailed presentation. For most honest athletes, these findings support the initiation, development, and refinement of strategies for identifying food supplements with added illegal substances. Every athlete must have access to these approaches in order to avoid becoming vulnerable to sports fraud.

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"Can We Trust Them?"- An Expert Evaluation of Large Language Models to Provide Sleep and Jet Lag Recommendations for Athletes. Sports Medicine, Sports Medicine, Ahead of Print: 1-14, 2025.

Background: With the increasing use of artificial intelligence in healthcare and sports science, large language models (LLMs) are being explored as tools for delivering personalized, evidence-based guidance to athletes.

Objective: This study evaluated the capabilities of LLMs (ChatGPT-3.5, ChatGPT-4, and Google Bard) to deliver evidence-based advice on sleep and jet lag for athletes.

Methods: Conducted in two phases between January and June 2024, the study first identified ten frequently asked questions on these topics with input from experts and LLMs. In the second phase, 20 experts (mean age 43.9 ± 9.0 years; ten females, ten males) assessed LLM responses using Google Forms surveys administered at two intervals (T1 and T2). Inter-rater reliability was evaluated using Fleiss' Kappa, and intra-rater agreement using the Jaccard Similarity Index (JSI), and content validity through the content validity ratio (CVR). Differences among LLMs were analyzed using Friedman and Chi-square tests.

Results: Experts' response rates were high (100% at T1 and 95% at T2). Inter-rater reliability was minimal (Fleiss' Kappa: 0.21-0.39), while intra-rater agreement was high, with 53% of experts achieving a JSI ≥ 0.75. ChatGPT-4 had the highest CVR for sleep (0.67) and was the only model with a valid CVR for jet lag (0.68). Google Bard showed the lowest CVR for jet lag (0%), with significant differences compared to ChatGPT-3.5 ($p = 0.0073$) and ChatGPT-4 ($p < 0.0001$). Reasons for inappropriate responses varied significantly for jet lag ($p < 0.0001$), with Google Bard criticized for insufficient information and frequent errors. ChatGPT-4 outperformed other models overall.

Conclusions: This study highlights the potential of LLMs, particularly ChatGPT-4, to provide evidence-based advice on sleep but underscores the need for improved accuracy and validation for jet lag recommendations.

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Characterizing body postures by injury scenarios - A video review analysis of hamstring strain injuries in the National Football League. BMJ Open Sport & Exercise Medicine, 11(4): e002749, 2025.

Objectives: The purpose of this study was to (1) establish the agreement of multiple expert reviewers' identification of hamstring strain injury (HSI) scenarios and related body postures among National Football League (NFL) players and (2) determine the prevalence of each HSI scenario and associated body posture in the NFL between 2018 and 2022.

Methods: Videos from 305 HSIs in the NFL were reviewed in a blinded fashion by two expert reviewers, who classified the injuries into seven predefined injury scenarios developed by a separate committee of experts. Lower extremity body postures were also identified. Cohen's Kappa coefficients were calculated to determine inter-rater agreement and used to select the subset of injuries to be described via injury scenario and body posture with the intent of minimizing bias or ambiguity in reporting.

Results: 137/305 (45%) injury videos met the criteria for inclusion in the final dataset based on assessment of classification agreement. Agreement in injury scenario and body posture ranged from poor to moderate. Sprinting injuries were the most common scenario (59/137, 43%). Unique scenarios specific to American football demonstrated the importance of excessive trunk flexion during contact, or during acceleration and change of direction (54/137, 39%).

Conclusions: Using a systematic approach involving multiple expert reviewers, sprint-related injury scenarios were highlighted as the most common injury scenario for HSIs. Specific to American football; however, excessive trunk flexion during contact plays or during acceleration and change of direction may be important to consider for injury prevention or rehabilitation from HSIs.

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Club-based vs hospital-guided rehabilitation after ACL reconstruction in football participants: A two-year prospective cohort study. Frontiers in Sports and Active Living, 7: 1641750, 2025.

Background: Anterior cruciate ligament (ACL) injury is a common and serious sports injury in football. Although reconstruction surgery can restore joint stability, postoperative rehabilitation strategies play a critical role in long-term outcomes, particularly return to sport (RTS). However, comparative studies on the long-term efficacy of hospital-guided vs. club-based rehabilitation programs remain limited.

Purpose: To compare the two-year clinical outcomes and RTS of hospital-guided and club-based rehabilitation programs following ACL reconstruction in football participants.

Methods: This prospective cohort study enrolled 40 football participants who underwent primary ACL reconstruction. Participants were assigned to two groups based on rehabilitation modality: hospital-guided rehabilitation (n = 20), which emphasized basic joint function recovery, and club-based rehabilitation (n = 20), which focused on sport-specific physical conditioning and performance enhancement. Outcome measures included subjective functional scores: Tegner Activity Score (TAS), Marx Activity Rating Scale (MARS), International Knee Documentation Committee (IKDC) subjective score, Lysholm Knee Rating Scale (LKSS), and Anterior Cruciate Ligament-Return to Sport after Injury (ACL-RSI), as well as objective knee function [effusion, range of motion (ROM), and joint stability], time to return to training and competition, and re-injury rate.

Results: The two groups were comparable at baseline. The club-based group demonstrated significantly better outcomes in TAS (8.5 ± 1.4 vs. 7.2 ± 1.0, p = 0.006), MARS (12.7 ± 2.8 vs. 10.6 ± 2.7, p = 0.019), and ACL-RSI (73.3 ± 21.4 vs. 54.3 ± 18.7, p = 0.003). Time to return to training and competitive play was significantly shorter in the club-based group. No significant differences were observed in ROM, joint stability, or re-injury rate between groups.

Conclusion: Club-based rehabilitation led to superior subjective function, greater psychological readiness, and earlier return to sport compared to hospital-guided programs, without increasing re-injury risk. These findings support the effectiveness of sport-specific rehabilitation in football players after ACL reconstruction, though further studies are needed to evaluate long-term safety and cost-effectiveness.

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Commotio cordis in sudden cardiac death in the young: A state-of-the-art-review. Reviews in Cardiovascular Medicine, 26(9): 43357, 2025.

Abstract - Commotio cordis is a rare but fatal cause of sudden cardiac death in young people, particularly athletes exposed to non-penetrating chest trauma. Commotio cordis occurs when an impact to the chest triggers a lethal ventricular arrhythmia in the absence of pre-existing structural heart disease. Despite advances in the understanding of commotio cordis, the prevention and management of this condition remain challenging. The literature indicates that commotio cordis is most common in adolescents and sports such as baseball, football, and ice hockey. The key pathogenic mechanism involves a chest impact occurring during a vulnerable phase of the cardiac cycle, leading to ventricular fibrillation. Immediate cardiopulmonary resuscitation and prompt use of an automated external defibrillator are crucial for survival. However, the effectiveness of preventive measures, such as chest protectors and greater awareness of cardiovascular emergencies, remains debated. As a leading cause of sudden death in young athletes, commotio cordis requires further research to refine prevention strategies and improve outcomes. This review provides an updated overview of the pathophysiological mechanisms, risk factors, intervention strategies, and preventive approaches for this condition.

Considering the effects of cannabinoids and exercise on the brain: A narrative review. Sports, 13: 320, 2025.

Abstract - Recently, there has been rising interest in the use of cannabis and its derivatives as therapeutic tools to support brain health, particularly in athletes. Cannabis-based substances interact with the endogenous cannabinoid (i.e., endocannabinoid) system, which is involved in widespread physiological processes that contribute to brain function. In other work, the benefits of exercise for brain health have been prominently noted. Despite large bodies of work on both cannabinoid and exercise influences on brain function, there is an understudied overlap in their physiological effects that may be especially important in athletic populations regularly engaged in high volumes of exercise. This narrative review describes mechanistic overlaps between cannabinoid and exercise effects on brain function. The literature search was broad, emphasizing research published since 2010 and including randomized clinical trials, observational studies, case studies, preclinical work, both human and animal

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studies, and information presented in related review articles. The focal point of the current review is the potentially overlapping effects of cannabinoids and exercise on brain function via physiological processes underpinning inflammation, vascular function, and neuroplasticity. Mechanisms are described in detail with consideration of common and contrasting influences of cannabinoids and exercise on the brain. Altogether, the compiled information suggests that indirect and direct interactions between these two therapeutic avenues have potential to introduce additive, synergistic, or opposing effects. Considering such interactions will be critical in optimizing therapeutic strategies involving cannabinoids as they are increasingly applied in the sport sciences and beyond.

Does the warmup really reduce injury risk? Why we need new approaches to confidently answer this question. Journal of Science and Medicine in Sport, Ahead of Print: 1-3, 2025.

Abstract - This viewpoint article challenges the common belief that warmups acutely reduce injury risk in sports. While coaches, players, and some studies (e.g., on programs like FIFA 11+ or SHRed Injuries Basketball) suggest warmups prevent injuries, the authors argue that existing research primarily examines the *chronic* effects of injury prevention programs (IPPs) implemented during warmups, not the *acute* effects of warming up itself. These studies often compare structured IPPs to "usual" warmups over time, conflating long-term fitness improvements with immediate warmup benefits.

The authors highlight a key issue: IPPs may reduce injuries due to repeated training stimuli, not warmup per se. For example, one study showed better results when FIFA 11+ was done both pre- and post-training, indicating effects beyond warming up.

To address this, they propose a refined study design: a volume-equalized chronic trial where both groups perform the same protocol, but one pre-training (as warmup) and the other post-training. This isolates warmup's acute role while controlling for total training dose. If the pre-training group has fewer injuries, it supports warmup's preventive effect; equal or higher rates would contradict it. The design applies to training sessions only, not matches, and is ethically sound as athletes naturally warm up during activities.

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In conclusion, current research designs fail to answer whether warmups truly prevent injuries acutely, and new approaches are needed to test this hypothesis rigorously. The article includes figures illustrating prevalent vs. proposed study designs.

Dose-response relationship between evening exercise and sleep. Nature Communications, 16: 3297, 2025.

Abstract - Public health guidelines recommend exercise as a key lifestyle intervention for promoting and maintaining healthy sleep function and reducing disease risk. However, strenuous evening exercise may disrupt sleep due to heightened sympathetic arousal. This study examines the association between strenuous evening exercise and objective sleep, using data from 14,689 physically active individuals who wore a biometric device during a one-year study interval (4,084,354 person-nights). Here we show later exercise timing and higher exercise strain are associated with delayed sleep onset, shorter sleep duration, lower sleep quality, higher nocturnal resting heart rate, and lower nocturnal heart rate variability. Regardless of strain, exercise bouts ending ≥4 hours before sleep onset are not associated with changes in sleep. Our results suggest evening exercise—particularly involving high exercise strain—may disrupt subsequent sleep and nocturnal autonomic function. Individuals aiming to improve sleep health may benefit from concluding exercise at least 4 hours before sleep onset or electing lighter strain exercises within this window.

Effects of post-exercise heat exposure on acute recovery and training-induced performance adaptations: A systematic review. Sports Medicine – Open, 11: 106, 2025.

Background: Whole-body heat exposure, such as sauna bathing or hot water immersion (HWI) has been shown to induce various physiological adaptations that can improve athletic performance. However, the effects of post-exercise heat exposure on acute recovery and promoting training-induced performance

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adaptations are not well understood. The aim of this systematic review was to summarise the current evidence on the effects of post-exercise heat exposure on physical performance in healthy adults.

Methods: This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A systematic search for articles was conducted in December 2023 and updated in June 2025 using the PubMed, SPOLIT, Medline, and SPORTDiscus databases. Eligible studies were randomised or crossover trials comparing whole-body post-exercise heat exposure (≥ 36 °C, e.g., hot water immersion to at least sternum level, or sauna bathing) to passive or placebo recovery. The risk of bias of the included studies was assessed using the Cochrane Collaboration Risk of Bias Tool version 2. Only studies that provided results on maximal physical performance outcomes in healthy adults were included.

Results: Fourteen studies, including a total of 194 participants, met the inclusion criteria. Nine studies investigated acute effects after heat exposure and five were long-term training intervention studies. The acute studies reported uncertain results, with studies showing no effects (n = 4), beneficial (n = 4), or adverse (n = 1) effects of post-exercise heat exposure on performance recovery. The chronic studies suggested that post-exercise heating may improve running performance, at least in hot conditions. However, repeated heat exposures had no effect on cycling performance or VO_2max. The overall quality of the evidence was low to moderate. The heterogeneity of study designs, heating protocols, exercise modes, performance outcomes and recovery times precluded meta-analysis.

Conclusion: Based on the current evidence, it is not possible to draw definitive conclusions about the effects of post-exercise heat exposure on recovery and physical performance development. Additional high-quality studies are needed to determine the optimal heat exposure methods and recovery strategies for different types of training and performance outcomes.

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Pro Football Sports Science Research Update

Ergogenic and physiological effects of sports supplements: Implications for advertising and consumer information. Nutrients, 17: 2706, 2025.

Background: The use of sports supplements has increased significantly in athletic contexts, raising the need to evaluate their efficacy, safety, regulatory status, and communication practices.

Objective: This study aimed to describe and synthesize the ergogenic and physiological effects of Australian Institute of Sport (AIS) Category A performance supplements.

Methods: A descriptive and observational study was conducted, collecting and analyzing information from systematic reviews and position statements related to performance supplements, including caffeine, creatine, β-alanine, nitrate/beetroot juice, sodium bicarbonate, and glycerol.

Results: Caffeine and creatine are the only supplements with authorized health claims. However, β-alanine, nitrates, sodium bicarbonate, and glycerol show positive ergogenic effects supported by strong evidence, especially in endurance, strength, high-intensity, and aquatic sports. However, these substances lack regulatory approval, and only a small proportion of commercial products comply with current legislation.

Conclusions: While performance supplements may enhance athletic performance when used alongside proper nutrition and scientific guidance, their effectiveness is not always consistent or assured. This review highlights the urgent need to update regulatory frameworks, harmonize labeling standards, and promote ethical marketing to safeguard consumers and support sports and nutrition professionals.

Foot strike angle as key predictor of multi-plane knee joint moments in sidestep cutting: Implications for ACL injury prevention. Sports Biomechanics, Ahead of Print: 1-12, 2025.

Abstract - Video analyses of ACL injury situations in change-of-direction tasks suggest an elevated injury risk during flatfooted or rearfoot landing. While loading in each individual plane of the knee can contribute to injury, combined loading seems to be the most likely mechanism. This study investigated if knee joint moments are related to the foot strike angle in pre-planned and unplanned handball-specific

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sidestep cuts performed by *n* = 51 female handball players. Canonical correlation analysis (CCA) with the foot strike angle at initial ground contact serving as the predictor variable was performed on the knee joint moment vector field (M_{xyz}). In both conditions, the foot strike angle was related to M_{xyz}. Subsequently, post-hoc tests were conducted using CCA for moment couples (M_{xy}, M_{xz}, M_{yz}) and linear regression for individual moment components. Results revealed that traditional analyses focusing on individual planes only partially explain the observed relationships, particularly during the unplanned cutting condition. Given the time periods in which correlations exist, this study shows that manipulating the foot strike angle has the potential to reduce multi-plane knee loading associated with ACL injury, while limiting analyses to individual planes might underestimate the potential to mitigate ACL injury risk.

Hamstring injury risk in male professional football (soccer): Do external training loads play a role? BMJ Open Sport & Exercise Medicine, 11: e002649, 2025.

Objectives: Hamstring strain injury (HSI) is the most common time-loss injury in football and is prone to recurrence. This exploratory study aimed to describe the relationship between short-term external training load (TL) and HSI occurrence in male senior football players at the professional level.

Methods: TL data in terms of global positioning system (GPS) variables and HSI incidence were collected over four seasons from 25 senior professional football players. GPS variables included total distance (TD), very intense accelerations (>3.00 m/s^2), very intense decelerations (<-3.00 m/s^2), high-speed running distance (>19.8 km/hour), maximal sprint distance (>29.8 km/hour) and maximal velocity. For each injury case, TL during the 7 and 14 days preceding the injury was compared with matched control periods from the same player.

Results: 25 hamstring injuries were included for analysis. Large interindividual variation in TL patterns was observed, with some players exhibiting higher and others lower TL in the periods preceding injury compared with control periods. No consistent group-level trends were identified between injured and control periods.

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Conclusion: This exploratory study found considerable individual variability in TL prior to HSI and no clear group-level patterns. These findings suggest that short-term TL metrics alone may have limited utility in predicting HSI risk at the group level in professional football players.

Impact of ankle injury history and pre-National Football League Draft training on lower limb coordination during running: A pilot study. Cureus, 17(10): e93887, 2025.

Objective: Ankle sprains are one of the most common injuries sustained by American football athletes. Players with a recent history of ankle sprain, even when asymptomatic, may have sensorimotor impairments that affect control of the hip and knee during dynamic activities such as speed running. Among players preparing for the National Football League (NFL) draft, these functional deficits have wide-ranging effects and warrant systematic investigation. This study aimed to determine the influence of a recent history of ankle sprain (≤6 months) and a six-week NFL draft preparation training camp on hip-ankle (HA) and knee-ankle (KA) coordination. Specifically, we examined HA and KA coordination angles and variability using vector coding. We hypothesized that the group with a recent ankle sprain history (RASH) would express less ankle dominant patterns, and these would improve post-training, and the RASH group would show impaired neuromuscular adaptability as evidenced by lower coordination variability, but this would improve post-training, through increased variability.

Materials and methods: This study analyzed a sample of 12 athletes with a recent ankle sprain history (RASH) and compared their running mechanics to 12 matched controls with no recent ankle sprain history (non-RASH). Running mechanics during a five-second 23 km/h run were evaluated pre- and post-training using a 10-camera motion capture system. Coordination was quantified into angles using vector coding, and coordination variability was quantified using circular statistics, with variability expressed as the standard deviation derived from the mean resultant length of angular data. Data for both were derived over four subphases of the running gait cycle: first half stance, second half stance, first half swing, and second half swing. A 2 (RASH and non-RASH) × 2 (pre and post) ANOVA were used to assess the influence of ankle sprain history and training on HA and KA coordination and coordination variability ($p \leq 0.05$).

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Results: No significant group × time interactions or main effects were found in any of the subphases for coordination angles or variability for HA and KA ($p > 0.05$). For HA, there were medium effect sizes in coordination angle for the second half of stance ($\eta^2_p = 0.075$) and coordination variability for the second half of swing ($\eta^2_p = 0.121$). The RASH group had larger increases in HA angle from training (212.7 ± 102.7° (pre) versus 235.5 ± 91.2° (post)) than the non-RASH group (268.7 ± 56.8° (pre) versus 265.5 ± 48.2° (post)). Further, the RASH group shifted from an in-phase hip-dominant to an in-phase ankle-dominant pattern in the second half of stance. For KA, there were medium effect sizes for the first half stance ($\eta^2_p = 0.093$) and second half swing ($\eta^2_p = 0.091$). Specifically, for the first half stance, the RASH group changed from anti-phase ankle to in-phase ankle control (97.0 ± 19.5° (pre) versus 85.5 ± 29.4° (post)), while the non-RASH group was stable at an anti-phase ankle pattern (93.6 ± 19.8° (pre) versus 95.6 ± 19.8° (post)). For the second half swing, both groups produced in-phase knee-dominant patterns, but the RASH group decreased their coordination angle by 4.4°, and the non-RASH group increased their coordination angle by 3.8°.

Conclusions: Highly skilled athletes with a recent history of ankle sprains adopt compensatory joint coupling strategies that persist even in the absence of symptoms and can be modified through training.

Impact of evening blue light exposure timing on sleep, motor, and cognitive performance in young athletes with intermediate chronotype. Biology of Sport, 42(3): 61-68, 2025.

Abstract - Evening exposure to blue light has been shown to negatively affect sleep patterns and cognitive function. However, the effect of the specific timing of blue light exposure on motor performance and sleep quality in adolescent athletes remains unclear. This study aimed to investigate the impact of evening blue light exposure (BLE) timing on sleep quality, motor performance, and cognitive function in young athletes with intermediate chronotypes. Sixteen male adolescent athletes (age: 15.11 ± 0.92 years, body height: 168.25 ± 7.09 cm, body-mass: 59.49 ± 5.63 kg) participated in a randomized, crossover study with four conditions: BLE from (i) 7: 30–9: 00 PM, (ii) 9: 00–10: 30 PM, (iii) 10: 30 PM-12: 00 AM, and (iv) control (no BLE). Sleep quality and quantity were evaluated using the Spiegel and Vis-Morgen questionnaires. Selective attention was assessed using the Bells Test, while

motor function was tested by dart-throwing accuracy and movement duration. ANOVA indicated substantial main effects of BLE timing on sleep metrics, motor performance, and cognitive function. Total sleep duration was markedly shorter in the 9: 00–10: 30 PM and 10: 30 PM-12: 00 AM conditions relative to the control ($p < 0.001$, $d = 0.54$ [medium] and $d = 0.87$ [large], respectively) and the 7: 30–9: 00 PM condition ($p < 0.05$, $d = 0.28$ [small] and $p < 0.001$, $d = 0.56$ [medium], respectively). Dart-throwing accuracy was significantly lower in the 9: 00–10: 30 PM and 10: 30 PM-12: 00 AM conditions versus the control ($p < 0.002$, $d = 0.77$ [medium]) and $p < 0.001$, $d = 1.41$ [large], respectively). Movement duration was significantly longer in these conditions compared to the control ($p < 0.001$, $d = 1.75$ [large] and $d = 1.51$ [large], respectively) and 7: 30–9: 00 PM condition ($p < 0.01$, $d = 1.38$ [large] and $p = 0.002$, $d = 1.17$ [large], respectively). Selective attention was significantly lower in the 9: 00–10: 30 PM and 10: 30 PM-12: 00 AM conditions compared to the control ($p = 0.003$, $d = 0.66$ [medium] and $p < 0.001$, $d = 0.91$ [large], respectively). Evening BLE, especially after 9: 00 PM, adversely affects sleep quality, motor performance, and cognitive function in young athletes. These findings underscore the necessity of reducing BLE in the evening to enhance sports performance, optimize training and recovery, and facilitate motor learning for skill development.

Implementing exposure limits for players in contact team sports: Review of principles and practices. Frontiers in Sports and Active Living, 7: 1593766, 2025.

Abstract - Concerns have been raised that professional athletes taking part in contact team sports, such as football and rugby union, are subject to the risk of post-career adverse health conditions. These health concerns include neurodegenerative diseases resulting from head impacts and osteoarthritis in lower limb joints due to wear and tear. There have been suggestions that athletes in contact team sports should be subject to exposure limitations to mitigate these risks. At the present time, little information or guidance is available for athletes and sport governing bodies about how such limitations should be identified and implemented. The criteria used for defining occupational health concerns and the role and nature of occupational exposure limits are discussed. Consideration is given to whether these criteria have been considered and embraced in research studies examining adverse health conditions in professional sport. Recommendations are presented for how future research studies investigating post-career, sport-related, adverse health concerns should be planned and implemented in

order to provide the occupational health information required to make evidenced-based decisions about potential health concerns in professional sport.

Is it all about the stars? Spending on star players and success in the National Football League. Omega: The International Journal of Management Science, Ahead of Print: 1-31, 2025.

Abstract - The National Football League (NFL) introduced a salary cap in 1994 which significantly affected the payroll efficiency of teams within the league. This paper uses a robust directional distance Data Envelopment Analysis (DEA) model to measure and evaluate the payroll efficiency of the 32 teams of the NFL from 2011- 2020. The focus is on investigating if teams are more payroll efficient when spending on star players or spreading their expenditure more equally across their squad. Four DEA-models were defined to analyze spending payroll efficiency relating to the overall allocation of salary, star offensive players, star defensive players, and starting quarterbacks. Inputs were relevant salary cap information, while outputs were performance statistics unique to each category. A relationship was discovered between payroll efficiency in each category and qualification for the playoffs and post-season success. Efficient expenditure on star defensive players was especially important.

Kinetics and kinematics of initial sprint acceleration performance in Division 1 collegiate American football players. Journal of Sports Science, Ahead of Print: 1-15, 2025.

Abstract - This study collected the first direct measurements of the kinetic and kinematic variables associated with initial acceleration performance in American football players. 123 male NCAA Division I American football players performed one-to-two maximal sprints of 20–30-meters. Kinetics were derived from a linear motorized encoder and individual linear sprint force-velocity-power (FVP) profiles were calculated utilizing a validated field method. Kinematics were derived from video images recorded in slow-motion. The first three steps, including the initial push-off, were used for analysis at the instants

of toe-off and touchdown. Large to nearly perfect, significant relationships were obtained between maximal power and 10-meter time. The spatiotemporal variables with the strongest relationships to maximal power were contact time, takeoff and step velocity, and normalized flight length. At toe-off, the swing leg hip angle, stance leg thigh angle, and CM-toe distance had large relationships with maximal power. At touchdown, the shank angles of both the stance and swing leg and the stance leg knee angle had large relationships. Maximal power associated with force output in the horizontal direction may considered a key performance indicator for initial acceleration in American football and certain technical underpinnings are associated with higher levels of maximal power during initial sprint acceleration

Mediterranean diet and athletic performance in elite and competitive athletes: A systematic review and meta-analysis. Nutrition, Metabolism and Cardiovascular Diseases, Ahead of Print: 1-15, 2025.

Aims: Mediterranean Diet (MedDiet) may represent the reference dietary pattern for athletes. Our aim was to study whether MedDiet could be associated with any changes in performance in adult athletes practicing different types of sport at various levels of competition.

Data synthesis: A systematic literature review (PROSPERO n. CRD42023459039) was performed based on the PRISMA method. The search was conducted in PubMed/Medline, Scopus, Web of Science, and the Cochrane Library in October 2024. Competitive/elite healthy adult athletes performing at least 6 h/week. The risk of bias was assessed using the RoB 2.0 Cochrane tool for randomized-controlled trials, Robins-I tool for non-randomized studies and the Newcastle Ottawa Scale for the observational studies. A meta-analysis was conducted. 9 studies were included in this review (192 participants, mainly males), 5 of them had a cross-sectional design. The sample size ranged from 10 to 43 athletes performing different sports. 5 studies reported an impact of MedDiet on athletic performance; of these, 4 studies reported a positive effect. MedDiet adherence was positively related to anaerobic/aerobic power, explosive strength, and indirectly associated with body fat percentage. Reported adherence to MedDiet varied from low to high. The meta-analysis showed no MedDiet effect on the performance outcomes [SMD 0.00 (CI -0.26; 0.25)].

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Conclusions: The findings of this review suggest a general positive influence of the MedDiet on athletic performance, supporting its role as a beneficial dietary pattern for athletes. Quantitative analysis did not confirm these results, probably due to the few studies included, their heterogeneity and low-to-moderate quality.

Motion capture technologies for athletic performance enhancement and injury risk assessment: A review for multi-sport organizations. Sensors, 25: 4384, 2025.

Background: Motion capture (MoCap) technologies have transformed athlete monitoring, yet athletic departments face complex decisions when selecting systems for multiple sports.

Methods: We conducted a narrative review of peer-reviewed studies (2015-2025) examining optical marker-based, inertial measurement unit (IMU) systems, including Global Navigation Satellite System (GNSS)-integrated systems, and markerless computer vision systems. Studies were evaluated for validated accuracy metrics across indoor court, aquatic, and outdoor field environments.

Results: Optical systems maintain sub-millimeter accuracy in controlled environments but face field limitations. IMU systems demonstrate an angular accuracy of 2-8° depending on movement complexity. Markerless systems show variable accuracy (sagittal: 3-15°, transverse: 3-57°). Environmental factors substantially impact system performance, with aquatic settings introducing an additional orientation error of 2° versus terrestrial applications. Outdoor environments challenge GNSS-based tracking (±0.3-3 m positional accuracy). Critical gaps include limited gender-specific validation and insufficient long-term reliability data.

Conclusions: This review proposes a tiered implementation framework combining foundation-level team monitoring with specialized assessment tools. This evidence-based approach guides the selection of technology aligned with organizational priorities, sport-specific requirements, and resource constraints.

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Next-Generation Approaches in Sports Medicine: The Role of Genetics, Omics, and Digital Health in Optimizing Athlete Performance and Longevity—A Narrative Review. Life, 15: 1023, 2025.

Abstract - This review aims to provide a comprehensive framework for implementing precision sports medicine, integrating genetics, pharmacogenomics, digital health solutions, and multi-omics data. Literature review was conducted using MEDLINE, EMBASE, Web of Science, and Cochrane Library databases (January 2018-April 2024), focusing on precision medicine applications in sports medicine, utilizing key terms including "precision medicine", "sports medicine", "genetics", and "multi-omics", with forward and backward citation tracking. The review identified key gene variants affecting athletic performance: endurance (*AMPD1*, *PPARGC1A*), power (*ACTN3*, *NOS3*), strength (*PPARG*), and injury susceptibility (*COL5A1*, *MMP3*), while also examining inherited conditions like cardiomyopathies (*MYH7*, *MYBPC3*). Pharmacogenomic guidelines were established for optimizing common sports medications, including NSAIDs (*CYP2C9*), opioids (*CYP2D6*), and cardiovascular drugs (*SLCO1B1*, *CYP2C19*). Digital health technologies, including wearables and predictive analytics, showed potential for enhanced athlete monitoring and injury prevention, while multi-omics approaches integrated various molecular data to understand exercise capacity and injury predisposition, enabling personalized assessments, training regimens, and therapeutic interventions based on individual biomolecular profiles. This review provides sports medicine professionals with a framework to deliver personalized care tailored to each athlete's unique profile, promising optimized performance, reduced injury risks, and improved recovery outcomes.

Peri-operative protein or amino acid supplementation for total joint arthroplasty: A systematic review and meta-analysis. Journal of Orthopaedic Surgery and Research, 20: 439, 2025.

Purpose: Osteoarthritis (OA) affects weight-bearing joints, such as hips and knees, and its prevalence is rising due to factors like obesity and aging. Muscle atrophy, exacerbated by aging and surgery, increases the risk of joint instability and falls. Orthopedic surgeons explore dietary interventions to counteract

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these effects, with protein supplementation (PS) showing promise. This systematic review and meta-analysis assessed the effectiveness of PS in arthroplasty patients, comparing findings with sports medicine and sarcopenia literature.

Methods: Following PRISMA guidelines, we searched PubMed, Web of Science, Scopus, and Embase (February 2025) for protein and amino acid supplementation studies in total knee or hip arthroplasty (TKA and THA) patients. The quality assessment used the Cochrane risk of bias and the Newcastle-Ottawa Scale. Meta-analysis calculated effect sizes for muscle atrophy and strength outcomes.

Results: Nineteen studies (903 patients) evaluated oral or intravenous protein/amino acid supplementation over a mean follow-up of 55.2 days. Essential amino acids (EAA) significantly reduced muscle atrophy in quadriceps femoris muscle mass (SMD: 0.69; 95% CI: 0.44 to 0.95) and hamstring muscle mass (SMD: 1.04; 95% CI: 0.52 to 1.55). However, effects on intramuscular adipose tissue (IMAT) and muscle thickness (MT) were inconsistent. Muscle strength outcomes varied, with no significant effect on quadriceps muscle strength (QMS) or handgrip strength (HGS). Intravenous amino acid infusion improved muscle protein synthesis and reduced perioperative blood loss.

Conclusions: Protein and amino acid supplementation can reduce muscle atrophy in hip or knee arthroplasty patients. While effects on muscle strength and function are mixed, intravenous supplementation offers benefits. Further standardized research is needed to confirm these findings.

Player-Level Tackle Training Interventions in Tackle-Collision Sports: A Systematic Scoping Review. Sports Medicine – Open, 11: 103, 2025.

Background: In tackle-collision sports, the tackle has the highest incidence, severity, and burden of injury. Head injuries and concussions during the tackle are a major concern within tackle-collision sports. To reduce concussion and head impact risk, evaluating optimal tackle techniques to inform tackle-related prevention strategies has been recommended. The purpose of this study was to perform a systematic scoping review of player-level tackle training intervention studies in all tackle-collision sports.

Methods: The Arksey and O'Malley's five-stage scoping review process and Levac et al.'s framework were used, along with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis

extension for Scoping Reviews (PRISMA-ScR) checklist. The main inclusion criteria were that the study included an intervention aimed at improving a player's tackle abilities, and the intervention had to be delivered/implemented at the player-level in a training setting.

Results: Thirteen studies were included in this review, seven studies in American Football (54%), followed by a combined cohort of rugby union and rugby league players (three studies; 23%), rugby union (two studies; 15%), and one study reported on a rugby league cohort (8%). Studies focused primarily on the tackler, with the intervention incorporating a form of instruction or feedback, delivered through video or an expert coach. Other interventions included an 8-week strength and power training program, designing practice sessions based on baseline data, and helmetless training in American Football. All interventions demonstrated a favorable change in the outcome measured-which included tackler and ball-carrier kinematics based on motion capture video, tackler proficiency scoring, tackling task analysis, head impact frequencies by xPatch head-impact sensor technology, head impact kinematics using head-impact sensors (helmet or skin patches) and football tackle kinematics with motion capture systems or video.

Conclusion: This review shows that a range of studies have been undertaken focusing on player-level training interventions. The quality of studies were rated as 'good', and all studies showed improvements in outcome measures. Coaches and policy makers should ensure tackle technique is profiled alongside other player characteristics, and an evidence-based approach to improving player tackling is adopted, improving both performance and reducing injury risk.

Key points: Only 13 studies tested or implemented interventions at the player level in tackle-collision sports. The focus of the studies was primarily on the tackler, with the interventions incorporating a form of instruction or feedback, which was delivered through video or an expert coach. Other interventions included an 8-week strength and power training program, designing practice sessions based on baseline data, and helmetless training in American Football. All interventions demonstrated a favorable change in the outcome measure and provide coaches and policymakers with tackle training insights.

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Practices and perceptions in hamstring training for injury prevention and enhancement of athletic performance: A qualitative analysis. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2025.

Abstract - Hamstring strain injuries (HSIs) remain a prevalent issue across sports, despite extensive research. Previously, the current authors have reported practices and perceptions of training for the mitigation of HSIs, however such practices are highly nuanced. In this study, the authors utilized semi-structured interviews to further understand applied practitioner perceptions and identify how evidence is translated into practice. Twelve practitioners (one physiotherapist and eleven strength and conditioning [S&C] coaches including one dual accredited S&C coach and physiotherapist) working in a range of sporting disciplines were interviewed across six broad categories of questioning (practitioner profile, job role challenges, resistance training, high-speed running training, athlete testing and monitoring and views on contemporary research). Thematic analyses highlighted key challenges relating to limited time spent with athletes, fixture schedules and planning of training across the broader multidisciplinary team (MDT) as well as strategies to deal with such challenges such as micro-dosing of the resistance training stimulus, utilization of isometric exercise during fixture congestion periods. However, analyses also highlighted some areas of research that are not well translated into practice such as recommendations for the use of low-volumes of the Nordic hamstring exercise to elicit positive adaptations in strength and repeated bout effects and minimal effective dosages for resistance training. The findings of the study highlight the need for continued athlete and practitioner education, better communication and planning in the MDT and further research into minimum effective dosages to elicit strength adaptation in elite level athletes.

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Proprioception and Central Nervous Excitability: Implications for Return-to-Play in Athletes with Ligament or Meniscal Injuries. International Journal of Sports Physical Therapy, 20(8): 1122-1130, 2025.

Background and purpose: While indicators such as joint stability, range of motion, and muscle strength are critical for assessing readiness to return to sports, athletes may still face challenges even when these benchmarks are met. Therefore, this study aimed to examine the relationship between sports injury history or return-to-play status and central nervous system excitability during muscle output adjustments, and to investigate the characteristics of evoked EMG during torque maintenance as a potential indicator for assessing readiness to return to sport.

Design: Case control, observational study.

Methods: Twenty-one female university basketball players participated, including athletes with and without a history of surgery. Participants were eligible if they had no current injuries. Those with a history of surgery had completed rehabilitation and were cleared for return to play. Subjects were instructed to maintain 25% of maximum effort of voluntary isometric contraction of knee extension and the torque, long-latency reflexes (LLR) were recorded. Torque maintenance was assessed under two conditions: visual feedback and verbal instructions. Frequency of LLR appearance, and an average of latency, duration, amplitude ratio, and number of phases were calculated. Comparisons were made using Wilcoxon signed-rank tests between the dominant and non-dominant legs, and in participants with a history of injury, comparisons were also made between the injured and non-injured legs.

Results: The visual feedback condition led to finer motor adjustments in uninjured knees compared to verbal feedback. For injured knees, the amplitude ratio of LLR was significantly higher with visual feedback, indicating increased central nervous system excitability. Athletes who had not returned to sports displayed greater torque fluctuation and lower LLR amplitudes, suggesting inadequate fine motor control.

Conclusion: These findings suggest that visual feedback enhances motor adjustments in athletes with previous injuries, indicating heightened reliance on central nervous functions.

Level of evidence: 3b.

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Redefining the draft pick valuation in the National Football League. Frontiers in Sports and Active Living, 7: 1628223, 2025.

Abstract - The National Football League (NFL) Draft plays a critical role in determining team compositions and enhancing the competitive balance within the league. This study examines the valuation dynamics of NFL Draft picks, contrasting traditional valuation paradigms like Jimmy Johnson's Pick Value Chart (PVC) to estimate the intrinsic value of draft selections. Utilizing metrics such as weighted approximate value (wAV), games played (GP), and seasons started (ST), we derived that late-round draft picks might be undervalued in the conventional PVC. Our regression results indicated that the final pick in the first round holds a value closer to 56% of the first pick's worth, contrasting sharply with PVC's 20%. Our refined model suggested that the value of a pick 200 is about 30 times more than what PVC projects. We incorporate round-specific "Traded Up" dummy variables into our player-level regression models to test whether trade-up selections yield better outcomes. Results reveal that early round trade-ups, particularly in Round 1, are associated with significantly lower player performance, suggesting inefficiencies driven by overconfidence. In contrast, trade-ups in Round 4 show a modest positive effect, while other rounds yield no consistent advantage. These findings indicate that teams may benefit more from accumulating late-round picks than from aggressively trading up early. This paper accentuates the need for NFL teams to integrate data-driven models for informed draft decisions, thereby deepening insights into NFL draft valuation.

Rethinking Acute Sports Injuries: Evidence for an Overuse Mechanism in Hamstring and ACL Injuries. Scandinavian Journal of Medicine & Science in Sports, 35: e70146, 2025.

Abstract - Sports injuries have traditionally been classified as acute or overuse based on their onset and associated circumstances. Hamstring strain injuries and anterior cruciate ligament (ACL) injuries are two common sports injuries that are typically implicitly considered to represent acute injuries. This brief review, however, argues that hamstring and ACL injuries may at least partly present as overuse injuries resulting from a mechanical fatigue phenomenon, rather than acute injuries. Human, animal, and cadaveric studies are discussed to support this view. For example, human studies show no kinematic

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deviation in the stride during which the hamstring injury occurs as compared to the preceding strides. Further, the location of injury and ultrastructural damage of hamstring injuries is largely comparable to that seen in repetitive muscle-tendon unit lengthening experiments in animals. For the ACL, repetitive simulated jump landings have been shown to lead to ACL failure despite the ACL load being well below its ultimate strength. Furthermore, analyses of ACL explants obtained from noncontact ACL-injured patients during reconstruction surgery indicate similar damage to cadaveric studies that repetitively loaded the ACL. In summary, studies with diverse methodological approaches support the view that mechanical fatigue may predispose hamstring and ACL tissues to failure at submaximal loads during seemingly normal movements. Although further research is needed to substantiate these hypotheses, recognizing mechanical fatigue as a factor in these injuries can inform training and rehabilitation protocols and open opportunities to use modeling approaches and wearable sensors to monitor tissue load and damage, ultimately reducing injury rates.

The Dose–Response Relationship of Exercise-Based Injury Prevention Programs: Implications for Research and Practice. Sports Medicine, Ahead of Print: 1-10, 2025.

Abstract - Exercise-based injury prevention programs (EIPPs) need a certain period of consistent practice to achieve the expected efficacy. Therefore, as with many other biomedical interventions, EIPPs can be characterized by a dose-response relationship. In this Current Opinion, we aim to present the concept of the dose-response relationship and explore the research and clinical implications stemming from it. First, we introduce its origins from pharmacology and define the terms dose and response in the context of EIPPs. Specifically, we describe how the dose is dependent on the duration of EIPP practice and how the response can be quantified from injury-related epidemiological outcomes. Then, we highlight that the vast majority of research studies examining the efficacy of EIPPs neglect their dose-response relationship, which can lead to a potential underestimation of the EIPP effect. To overcome this problem, we present two statistical approaches that were applied to a previous study, providing a practical way to estimate the dose-response relationship of an EIPP. Finally, we discuss how researchers can integrate this concept in their study designs and analyses and how practitioners can plan the

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implementation of EIPPs on the basis of the time needed for the EIPPs to gain efficacy according to the dose-response relationship.

The Impact of Injury Severity on Sleep Quality in Football (Soccer) Players. European Journal of Sport Science, 4(3): 1-8, 2025.

Abstract - Soccer players are particularly vulnerable to musculoskeletal injuries, which not only impact their performance but also significantly affect sleep quality. Pain, inflammation, and psychological stress from injuries frequently lead to sleep disturbances, disrupting essential recovery processes. Moreover, poor sleep can increase the risk of reinjury, creating a cycle that hampers long-term athletic performance. This review synthesizes existing literature on the complex relationship between injury severity and sleep quality in soccer players. It explores how pain and inflammation contribute to sleep disturbances, how disrupted sleep alters recovery mechanisms, and how psychological stress exacerbates both injury outcomes and sleep disruptions. Additionally, it examines the long-term effects of poor sleep on recovery rates and reinjury risk, emphasizing its impact on career longevity. To address these challenges, this review highlights practical strategies for optimizing sleep, including effective pain management, improved sleep hygiene, relaxation techniques, and targeted nutritional interventions. The integration of sleep monitoring tools and individualized approaches can help coaches, trainers, and medical professionals prioritize sleep as a fundamental component of injury recovery. By acknowledging sleep as a key factor in both injury prevention and rehabilitation, this review underscores the need for a holistic approach to athlete care. Enhancing sleep quality can significantly improve recovery outcomes, reduce reinjury risks, and ultimately support sustained performance and well-being in soccer players.

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The Role of Vitamin D in Postoperative Tendon Healing: A Scoping Review. The Orthopaedic Journal of Sports Medicine, 13(10): 1-9, 2025.

Background: Tendon injuries pose significant clinical challenges because of poor vascularity and complex biomechanics, often leading to suboptimal healing. While surgical advancements and rehabilitation protocols have improved outcomes, delayed healing and high retear rates remain prevalent. Vitamin D, a key regulator of musculoskeletal health, may influence tendon healing through extracellular matrix remodeling, inflammation modulation, and cell proliferation. However, its effect on tendon repair remains unclear, with most studies focusing on rotator cuff tendons. This scoping review synthesizes current evidence on vitamin D's role in tendon healing and identifies gaps in the literature.

Purpose/hypothesis: The purpose of this scoping review is to evaluate the available literature on vitamin D's role in tendon healing and its influence on postoperative recovery. It was hypothesized that vitamin D deficiency is associated with poorer healing outcomes and increased failure rates.

Study design: Scoping review; Level of evidence, 4.

Methods: A systematic search of Embase, OVID Medline, and Emcare databases was conducted through November 2024. Eligible studies included observational studies and randomized controlled trials assessing vitamin D status in adults undergoing tendon repair. Two independent reviewers screened and extracted data, resolving discrepancies with a third reviewer. Study quality was evaluated using the methodological index for non-randomized studies score. Given the heterogeneity of studies, results were synthesized narratively, focusing on evidence gaps across different tendons.

Results: A total of 10 studies met inclusion criteria, with the majority focusing on rotator cuff repairs (n = 8). Vitamin D deficiency was consistently associated with poorer postoperative outcomes, including delayed healing, higher retear rates, and decreased functional recovery. Mechanistic studies suggested that vitamin D enhances healing through extracellular matrix remodeling and inflammation modulation. However, the overall quality of evidence was low, with most studies being retrospective. Notably, half of the studies relied on the same large administrative database, leading to potential patient overlap and limiting independent conclusions. Research on vitamin D's role in healing other tendons, such as the Achilles, patellar, and flexor tendons, was lacking.

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Pro Football Sports Science Research Update

Conclusion: Although evidence suggests vitamin D may support tendon healing, particularly in rotator cuff repair, studies remain low quality and limited to a single tendon group. Future prospective cohort studies and randomized controlled trials are needed to establish causality, assess its effects across various tendons, and determine optimal supplementation strategies.

Uphill Walking Unveils Hidden Instability and Biomechanical Adaptations in Anterior Cruciate Ligament-Deficient Copers. Gait & Posture, Ahead of Print: 1-26, 2025.

Objective: Using computerized motion analysis, the study investigated possible joint instability and biomechanical adaptations during uphill walking in participants with anterior cruciate ligament deficiency (ACLD) classified as copers based on level walking performance.

Methods: Thirteen participants with ACLD and thirteen healthy adults each walked and ascended a 3-meter slope with inclines of 3°, 6°, and 9°. Trunk angular motions, lower-limb joint angles, forces and moments, and tibial inclination angles were calculated. Two-way analyses of variance were used to study between-subject (group) and within-subject (slope angle) effects on the variables, using a significant level of $\alpha = 0.05$ for all tests.

Results: The ACLD group displayed a compensatory trunk anterior tilt strategy over the stance phase of uphill walking to reduce knee extensor moments and posterior knee forces to accommodate knee instability during uphill walking. Compared to Control, the ACLD group showed increased anterior forces during single-limb support (SLS) and decreased posterior forces at the affected knee during the double-limb support (DLS) of uphill walking. They also demonstrated an increased anterior tibial angle of the affected limb with a decreased extensor moment at the affected knee around the contralateral heel-strike. The trunk anterior tilt was also accompanied by increased extensor moments at both hips and ankle dorsiflexion during terminal DLS.

Conclusions: This study demonstrates that uphill walking uncovers hidden functional instability at the ACL-deficient knee that remains undetected during level walking. This suggests that assessments based solely on level walking may overestimate the functional stability of the ACLD knee during more

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demanding uphill walking. Incorporating uphill walking into clinical evaluations may help identify compromised movement patterns that could necessitate surgical intervention in ACLD.

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